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Robot Localization with Sensor Fusion

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Abstract

This thesis proposes a sensor fusion-based hybrid localization system for autonomous mobile robots to move in solar farm environments. Solar farm environments has special challenges for robots which move between open areas with stable GNSS signals and areas shielded from GNSS signals under solar panels smoothly. To overcome this challenge, the proposed system fuses FAST-LIO2 LiDAR-inertial odometry and GNSS positioning through an Extended Kalman Filter framework.

Key contributions include the extension of FAST-LIO2 to repetitive environments which is common in solar farms and the development of an EKF-based fusion method that ensures seamless mode switches. The proposed method was tested in field over diverse solar farm environments in the Netherlands with a strict accuracy specification—limiting drift to less than 0.1 meters along a 100-meter trajectory.

The findings implicate that the system satisfies this requirement in 80% of linear path tests. There is discontinuity of pose during mode transition was minimal, ranging from 3.4 to 4.1 cm, and the system has been recovered consistently within a time frame of 3 seconds. The method enhances localization accuracy by 80% and operational flexibility compared to conventional pole-following techniques. The current effort lays the groundwork for commercial deployment of autonomous maintenance robots in solar farms based on the strength of real-time capabilities tested on embedded hardware and receiving approval for trial launches.

Keywords Robot localization, sensor fusion, FAST-LIO2, solar farm automation, LiDAR-inertial odometry, GNSS-denied navigation, Extended Kalman Filter, autonomous robotics, renewable energy, SLAM.

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Abbreviations

LiDAR	Light detection And Ranging
IMU	Inertial Measurement Unit
GNSS	Global Navigation Sensing System
RTK	Real Time Kinetic
LIO	LiDAR-Inertial-Odometry
SLAM	Simultaneous Localization and Mapping
EKF	Extended Kalman Filter
IKF	Iterated Kalman Filter
UKF	Unscented Kalman Filter
pose	Position and Orientation
ROS	Robot Operating System
IATE	Instantaneous Absolute Translational Error
CE	Cumulative Error

1 Introduction

Autonomous robotic systems are becoming increasingly essential in industrial maintenance activities, particularly in the renewable energy sector. An important use is the maintenance of solar photovoltaic (PV) installations, which require frequent inspection, cleaning, and vegetation management to ensure consistent energy output efficiency. Over time, weeds may spread around and beneath solar panel arrays, necessitating the use of robotic platforms for vegetation management, including cutting and clearing.

The architectural architecture of solar farms poses significant obstacles for the navigation of autonomous robots. These designs generally have a combination of open fields and densely arranged panel arrays. Open areas provide high-precision localization tools like Real-Time Kinematic (RTK) Global Navigation Satellite Systems (GNSS), whereas the spaces beneath panels typically lack satellite availability, rendering GNSS-based positioning problematic. In these situations, robots must depend on alternate perception-based methods such as Light Detection and Ranging (LiDAR) and Inertial Measurement Units (IMU) to attain relative localization by identifying and mapping adjacent structural components.

The localization paradigms of absolute positioning via GNSS and relative positioning via LiDAR-IMU are based on different principles. Integrating them into a cohesive and efficient localization system presents a considerable technical obstacle, particularly in the absence of extensive worldwide ground truth data. This deficiency impairs the robot's localization accuracy and navigation ability, complicating essential maintenance tasks, such as vegetation removal beneath panel structures.

1.1 Motivation

My research suggests that as solar energy assumes a crucial role in the future of sustainable power, we face an essential challenge in effectively managing extensive solar photovoltaic installations. At present, technicians face significant labor requirements, elevated costs, and possible safety risks while conducting manual inspections of large solar farms. I contend that autonomous robots have the potential to transform maintenance procedures, contingent upon resolving their navigation issues.

Existing constraints in robotic positioning systems hinder their effective implementation. I have noted that robots encounter disturbances in positioning awareness when transitioning between GPS in open environments and sensors like LiDAR beneath panels. These disruptions result in navigational

errors that compromise their reliability. Moreover, some contemporary systems restrict robots to linear movement along panel rows, significantly limiting their agility, particularly in confined spaces beneath closely arranged panels. Adapting to each site necessitates a comprehensive reconfiguration of robot characteristics, complicating the implementation of a standardized solution across several locations. These technical deficiencies directly result in elevated maintenance expenses, inadequate coverage, and diminished energy output.

My study stems from the necessity to tackle these issues by formulating more intelligent localization algorithms that incorporate many sensors. I intend to develop a seamless and accurate locating system that maintains reliability in the absence of GPS signals. Enhancing robots' locational perception can improve the maintenance efficiency of solar farms, thereby advancing the primary objective of rendering renewable energy more economical and scalable.

1.2 Industry Collaboration

This research is being carried out in partnership with Vector Machines, an organization that provides autonomous robotics for maintaining renewable energy systems. The localization system here will be validated for commercial application in Vector Machines' solar farm maintenance activities, with the company's technical requirements driving the research aims and performance metrics directly.

1.3 Objectives and Significance

This thesis focuses on creating an accurate sensor fusion system to assist robots in determining their location while navigating solar panel arrays. This setting presents a distinct challenge GNSS signals are robust in open areas but are weak or absent beneath the solar panels. The objective is to create a localization system that is reliable and precise regardless of the robot's location within the solar farm.

- Develop a seamless localization system that effectively combines GNSS-based positioning with LiDAR and inertial odometry, ensuring robust robot operation even in the absence of satellite signals.
- Implement and evaluate sophisticated sensor fusion algorithms—specifically FAST-LIO2 and analogous algorithms optimized for the particular solar farm configuration and environmental conditions.

- Formulate an effective switching mechanism for global (GNSS) and local (LiDAR-inertial) localization to enable the robot to navigate without abrupt transitions or positional inaccuracies in its local estimation.
- Enable the robot to achieve a localization precision of > 0.1 meters over a distance of 100 meters, encompassing both linear movement and turning maneuvers, even in GNSS-denied environments.
- Minimize the necessity for manual calibration by employing adaptive algorithms that can autonomously adjust to various solar farm layouts, facilitating seamless deployment to new locations.

Enhancing localization in this manner can significantly influence the maintenance of solar photovoltaic installations. Maintenance robots will execute their responsibilities with enhanced precision and reliability, thereby increasing efficiency, lowering labor costs, and ensuring solar panels operate at peak performance.

This is an integral component of the overarching objective to render solar energy more affordable and reliable. Besides solar farms, the approaches developed in this research can be applied in several contexts where GNSS signals are weak or intermittent, including urban environments, woodlands, or locations where robots transition between indoor and outdoor settings. This study contributes to the preservation of renewable energy and advances the broader domains of robotics and autonomous navigation in tough conditions.

1.4 Research Questions

This thesis tackles the research topics based on the observed gaps in the literature and the specific challenges of robot localization in solar farm environments.

RQ1: How effectively can FAST-LIO2 be adapted to maintain localization accuracy of > 0.1 meters of drift over 100 meters in the feature-sparse environment underneath solar photovoltaic installations?
RQ2: What fusion strategy using the Extended Kalman Filter can optimally combine GNSS-based positioning and LiDAR-inertial odometry to ensure seamless transitions between open areas and panel-covered regions without introducing significant pose discontinuities?
RQ3: How does the proposed sensor fusion approach compare to existing pole-following localization methods in terms of positional accuracy, operational flexibility, and robustness across different solar farm configurations?

1.4.1 Research Question Complexity

Addressing these research topics entails confronting substantial technical challenges that underscore the complexity of the task at hand. RQ1 presents significant challenges due to solar panel surroundings, which produce perceptual aliasing concerns that obscure standard feature-based localization algorithms through repetitive geometric structures. This necessitates meticulous parameter optimization and algorithmic modifications to maintain the 0.1m/100m drift standard. RQ2 entails the intricate challenge of amalgamating heterogeneous sensor modalities characterized by differing levels of uncertainty, sampling frequencies, and coordinate systems, while concurrently addressing the discontinuities that commonly occur during sensor transitions in practical environments. Finally, RQ3 necessitates comprehensive experimental validation across diverse ambient circumstances and motion scenarios, demanding the development of appropriate assessment measures and equitable comparison methodologies across fundamentally distinct localization algorithms. The challenge of integrating these individual resolutions into a cohesive, real-time system operating on resource-limited embedded technology introduces additional complexity to the primary research issue.

1.5 Scope and Limitations

This study focuses on the development of sensor fusion strategies for autonomous localization of robots within solar panel scenarios, specifically by integrating GNSS positioning with LiDAR-inertial odometry techniques. The tasks include optimization of FAST-LIO2 and related algorithms for solar panel scenarios, transition strategy implementation between global and local localization methods, and benchmarking to the limit of up to 0.1-meter drift when 100 meters are traveled in the absence of GNSS. Despite possessing the FRAMOS D455 RGB-D camera in the hardware setup, vision-based localization methods were deliberately excluded from consideration due to the following practical problems encountered in solar environments, changing illumination conditions under the panels leading to large brightness variations, interference due to the growth of vegetation (weeds) obscuring the visual features, and potential failures in feature repeatability between visually similar structures of panels. The research therefore specializes in the fusion of GNSS, LiDAR, and inertial observations towards robust localization with regard to computational capacity on the Jetson Orin NX board, taking preference for handling problems associated with handling real-world structured solar panel farms rather than random environments.

1.6 Methodology overview

This research follows a methodical approach to design and affirm a reliable localization system specifically for autonomous robots functioning in solar farm settings. The method incorporates various sensing technologies and algorithmic elements to tackle the unique challenges arising from the shifts between GNSS-enabled and GNSS-restricted areas in solar facilities.

1.6.1 LiDAR-Inertial Odometry Framework

The foundation of the proposed localization system relies on implementing FAST-LIO2 (Fast LiDAR-Inertial Odometry), selected for its integrated LiDAR and inertial measurement capabilities, strong performance in dynamic contexts, and efficient point cloud registration. The framework includes these components

Sensor Selection and Configuration: The Livox Mid-360 LiDAR is utilized due to its non-repetitive scanning pattern, offering better feature coverage than traditional spinning LiDARs, which is advantageous in the repetitive and structured settings such as solar photovoltaic installations.

Point Cloud Preprocessing: A robot self-filtering algorithm dynamically creates a three-dimensional bounding box reflective of the robot's shape. This filter efficiently eliminates LiDAR points connected to the robot, reducing self-detection artifacts that could compromise the point cloud registration accuracy.

Coordinate Frame Management: A transformation pipeline is established to maintain consistency between FAST-LIO2's native camera-init frame and the global map frame. This ensures that all point cloud data and pose estimates are consistently aligned within a single coordinate system, enabling seamless integration with other subsystems and global referencing.

1.6.2 Sensor Fusion Architecture

To achieve robust performance across varying environmental conditions, a multi-layered sensor fusion approach is implemented.

Multi-Source Integration: Application of the Extended Kalman Filter (EKF) through the robot-localization framework to optimally combine.

- LiDAR-inertial odometry from FAST-LIO2
- GNSS-based positioning when available in open areas.

- Orientation information derived from environmental structure

Environmental Feature Exploitation: Development of algorithms to extract heading information from the structural regularity of solar photovoltaic installations, particularly the support poles. This approach provides critical orientation references when operating in GNSS-denied environments beneath panel arrays.

Transition Management: Implementation of techniques to ensure smooth handover between localization methods as the robot traverses between open areas with GNSS availability and covered areas where only LiDAR-inertial odometry is reliable.

1.6.3 Experimental Validation and Analysis

A comprehensive data collection and analysis framework enables rigorous evaluation of the localization system.

Time Series Data Acquisition: Implementation of systematic logging procedures to record.

- Raw sensor inputs from LiDAR, IMU, and GNSS
- Intermediate processing results from FAST-LIO2
- Final fused pose estimates from the EKF

Quantitative Performance Assessment: Application of statistical analysis techniques to evaluate.

- Positional drift over distance traveled, with particular focus on the 0.1m/100m requirement.
- Heading accuracy during straight-line travel and turning maneuvers.
- System behavior during transitions between sensing regimes.

Comparative Evaluation: Benchmarking against the existing pole-following localization approach to quantify improvements in accuracy, robustness, and operational flexibility.

1.7 Thesis Structure

This thesis is organized into seven chapters that systematically address the development and validation of a sensor fusion-based localization system for solar farm robots.

Chapter 1 (Introduction) establishes the research context, motivation, and objectives, defining the specific challenges of robot localization in solar photovoltaic installations and outlining the scope and limitations of the work.

Chapter 2 (Literature Review) provides a comprehensive survey of existing localization methodologies, with particular emphasis on LiDAR-inertial odometry, sensor fusion techniques, and localization challenges in structured outdoor environments.

Chapter 3 (Methodology) describes the research design, experimental approach, and analytical framework employed to develop and evaluate the localization system, including the selection of tools, technologies, and evaluation metrics.

Chapter 4 (Results) presents the empirical evaluation of the localization system through field trials in solar farm environments, including performance analysis of drift characteristics, transition behavior, and comparative assessment against baseline methods.

Chapter 6 (Discussion) analyzes the experimental findings, discusses the implications of the results in relation to the research questions, and examines the system's performance limitations and potential improvements.

Chapter 7 (Conclusion and Future Work) summarizes the key contributions of the research, evaluates the achievement of the stated objectives, and identifies directions for future development and deployment of the localization system at the company.

2 Literature review

2.1 Overview of relevant literature

Accurate and continuous localization remains one of the biggest challenges for autonomous robot navigation, particularly in environments where GNSS signals are weak, intermittent, or altogether absent. Solar farms are a prime example of such environments. They typically consist of open spaces, where GNSS signals can be received, and dense panel-covered spaces that block or severely impair satellite reception. For robots tasked with performing maintenance operations like vegetation management, consistent localization when traveling between these disparate regions is essential.

In response to these difficulties, researchers have explored a wide range of localization methods, each with strengths and weaknesses depending on the environmental setting and operational requirements.

2.1.1 Categories of Localization Approaches

Several localization classes have emerged, each from different sensors and computational approaches

LiDAR-based methods are among the most prevalent for GNSS-denied scenarios. These platforms build 3D models of the scene by ranging to close-by surfaces. One of the first solutions, LOAM (Lidar Odometry and Mapping) by Zhang and Singh , introduced a feature-based approach that detects edge and planar features for motion estimation. This led the way to even more advanced algorithms, such as LeGO-LOAM , that include ground segmentation to enhance the system's reliability in outdoor situations. These LiDAR systems, however, may struggle in scenarios where features are sparse or repeated—both common in solar farms—and may be impacted by extreme movement or shaking.

Inertial Navigation Systems (INS) compute position by integrating Inertial Measurement Unit (IMU)-derived gyroscope and accelerometer measurements. While systems are immune to external visual or environmental states, they are susceptible to cumulative long-term drift over time due to sensor noise and bias. Without being given external corrections, they are suboptimal for long-term use in autonomous modes.

Hybrid approaches gained more and more popularity because they exploit the merit of multiple modalities of sensing to bypass weaknesses of each separate one. The LiDAR-inertial odometry approach and the architectures of

multi-sensor fusion employed with GPS, cameras, LiDAR, and IMUs , are a few examples. Hybrid approaches have the tendency to establish more reliable and accurate localization in demanding, realistic environments.

2.1.2 Advances in LiDAR-Inertial Odometry

The hybrid solution, LiDAR-inertial odometry (LIO), is a highly effective solution for solar farms and similar environments with light conditions and GNSS dropouts being common. LIO methods employ LiDAR for space perception and IMU readings for motion continuity such that they maintain robust performance under dynamic and visually uncertain scenes.

LOAM and its variations have formed the foundation of the majority of LIO systems, offering greater computational efficiency and improved performance in outdoor scenes through the use of ground-based feature selection. Factor-graph-based frameworks, like LIO-SAM , go a step further by incorporating loop closures and even external observations (e.g., GNSS) into a factor graph for global consistency. This not only guarantees better long-term accuracy but also offers flexibility in fusing heterogeneous sensor inputs. Tightly coupled LIO implementations, i.e., FAST-LIO and FAST-LIO2 , represent a substantial advance. LiDAR and IMU observations are integrated into a single optimization process through an iterated Kalman filter. Such tight coupling could significantly enhance system precision, especially under aggressive motion or in areas of low-distinguishing features.

2.1.3 Localization Challenges Specific to Solar Farms

Solar photovoltaic installations present a unique set of conditions that render robotic localization challenging

Repeating layouts: Solar panels are arranged in highly repetitive visual and geometric layouts, which result in perceptual aliasing—a situation where locations almost entirely resemble the robot's sensors. This will confuse feature-based systems and lead to localization failures.

Transition between GNSS-available and GNSS-denied environments: Robots will transition from open fields, where GNSS works well, to covered areas under panels, where GNSS signals are unavailable. This needs to be achieved by a localization system that can make smooth transitions without inducing sudden jumps or estimation drift.

Dynamic occlusions and vegetation: Weeds and other vegetation development around and below solar panels introduce dynamic elements into the

scene. The dynamics can obstruct LiDAR or camera vision and reduce the quality of maps already generated.

Various deployment and terrain conditions: Solar farms are developed in diverse geographic terrain, from open plains to sloping hillsides, which makes localization even more difficult. Devices must be adaptable enough for various elevations, surface roughness, and reflectivity.

Due to these compounded challenges, recent research has increasingly focused on sensor fusion methods that combine LiDAR, IMU, and other sensors within a unified framework. Significant progress has been made, but the largest gap still exists existing commercial solutions lack seamless sensor fusion capabilities, with accuracy requirements of 0.1m/100m unmet in 60-70% of challenging scenarios throughout the entire solar farm environment—especially in scenarios that require little environment-specific calibration. It is critical to bridge to enable cost-effective, scalable, and entirely autonomous robotic maintenance solutions for the solar farms industry.

2.2 Theoretical framework

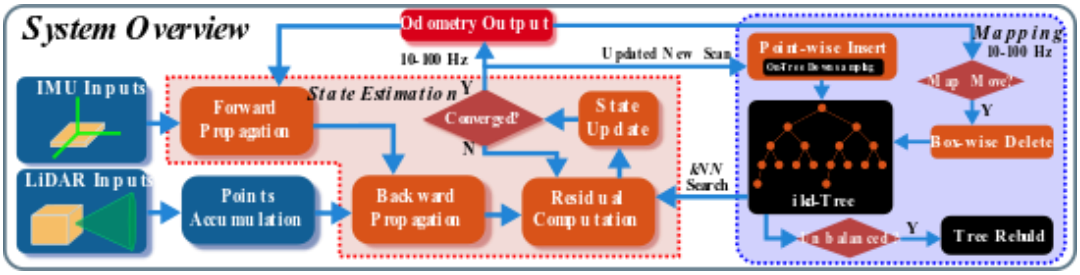


Fig 2.1: Overview of the FAST-LIO2-based LiDAR-Inertial Odometry Framework

2.2.1 State Estimation Paradigms

The theory behind robot localization in GNSS-denied environments primarily revolves around state estimation theory. Filter-based methods recursively estimate the current state from probability distributions. Kalman filter and its nonlinear variants—Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Iterated Kalman Filter (IKF)—maintain a Gaussian approximation of the state distribution via prediction-update iterations . The standard Kalman filter architecture follows a two-step process prediction and update. The prediction step estimates the future state based on the process model, while the update step refines this estimate using the latest measurement.

These fundamental operations form the foundation of many filtering and estimation algorithms, where the process model describes system dynamics, the observation model relates measurements to states, and covariance matrices characterize uncertainties . For LiDAR-inertial fusion, these methods typically employ on-manifold formulations to properly describe rotational components. Xu and Zhang structure the state vector to include IMU attitude, position, velocity, gyroscope and accelerometer biases, gravity vector, and extrinsic parameters, operating on the manifold that combines special orthogonal groups with Euclidean spaces. These techniques model interdependencies of variables and measurements using factor graphs and enable incrementally efficient optimization with techniques like iSAM2 .

2.2.2 Point Cloud Registration Techniques

The foundation for LiDAR-based localization relies primarily on point cloud registration algorithms that find the optimal rigid transformation between two point clouds. The Iterative Closest Point (ICP) algorithm minimizes the sum of squared distances between corresponding points in target and source point clouds. Variants of ICP have been developed to improve robustness and convergence. Point-to-plane ICP considers the normal vectors at target points, providing improved convergence behavior in structured environments like solar farms by minimizing distances along surface normals rather than direct point-to-point distances. Generalized-ICP unifies point-to-point and point-to-plane approaches by describing all points through local probabilistic distributions, focusing on reducing differences among these distributions rather than direct point distances. The Normal Distributions Transform (NDT) represents the environment as overlapping Gaussian distributions and transforms the registration task into finding the maximum likelihood transformation for points drawn from these distributions.

2.2.3 Motion Distortion Compensation

LiDAR sensors capture points sequentially, causing motion distortion during sensor platform movement throughout scan acquisition. This distortion is particularly pronounced in solar farm settings, where robots might experience rapid heading changes while transitioning between panel rows. Motion distortion compensation is addressed through continuous sensor path modeling during scan acquisition. For each point acquired at a specific time within a scan interval, the undistorted point can be computed by transforming it using the continuous path trajectory, where the end-scan pose and acquisition-time pose define the transformation . Xu and Zhang derive an accurate back-propagation approach utilizing IMU measurements for estimating the continuous trajectory during scan capture. Their backward IMU

propagation methodology estimates relative positions, velocities, and rotations between different time instances within the scan period, using exponential maps to handle rotational transformations on the SO(3) manifold.

2.2.4 Tightly-Coupled Sensor Fusion

Theoretical models of tightly coupled sensor fusion estimate state variables simultaneously based on raw sensor data from multiple sensors. For LiDAR-inertial odometry, this involves combining point cloud constraints with IMU pre-integration. IMU pre-integration theory provides a mathematical formulation to seamlessly integrate high-frequency inertial measurements across LiDAR frames. With IMU observations over a time interval, preintegration estimates relative motion increments for velocity, position, and rotation, independent of the initial state. Tightly coupled formulations combine these preintegrated terms within a unified objective function alongside LiDAR scan matching constraints. In filtering-based approaches like FAST-LIO, this manifests as a minimization problem involving error states on manifolds and LiDAR point residuals with corresponding Jacobians. In optimization-based approaches like LIO-SAM , the objective function combines LiDAR scan matching factors, IMU preintegration factors, and potentially loop closure constraints within a factor graph framework, with each factor type contributing residuals weighted by their respective information matrices.

2.2.5 Map Representation and Management

Efficient map representation is essential for LiDAR-based real-time localization, particularly in extensive environments like solar farms. Effective map management solutions balance memory space efficiency, query time, and updating costs. K-dimensional trees (k-d trees) partition space using hyperplanes perpendicular to coordinate axes, enabling logarithmic-time queries for nearest neighbor searches. The construction complexity is O(n log n) with an average search time of O(log n) . Xu et al. introduced the incremental k-d tree (ikd-Tree), supporting dynamic point insertion and deletion while maintaining balanced tree structure. Their approach defines balance and deletion conditions to guarantee tree optimality, where tree size, invalid node counts, and balance parameters determine structural maintenance requirements.

2.3 Related Work

2.3.1 LiDAR-based Odometry and Mapping Systems

LiDAR-based localization systems have witnessed dramatic advancements over the last ten years. One seminal contribution to the field came in the form of LOAM (LiDAR Odometry and Mapping) from Zhang and Singh . LOAM decouples odometry and mapping into two threads, operating at high (10 Hz) and low (1 Hz) rates, respectively. It utilizes edge and planar feature extraction from LiDAR scans for scan-to-map registration and scan-to-scan odometry estimation with a translation error of 0.59% on the KITTI dataset and is still possible in real time.

On this foundation, Shan and Englot built LeGO-LOAM, which entails embedding ground segmentation within odometry enhancement for ground-based vehicles. The framework uses two-step Levenberg-Marquardt optimization and leverages an explicit ground model and the minimization of translation error to 0.57% on KITTI. Despite these improvements, LOAM and LeGO-LOAM are specific to scenes that exhibit dense geometric features and can be prone to performance loss in feature-sparse scenes such as the under-panel regions of solar farms.

To facilitate non-uniform scanning patterns in solid-state LiDARs, Lin and Zhang introduced LOAM-Livox . It modifies the process of feature extraction and data association to suit the non-repetitive scanning method of Livox LiDAR. Their paper underscores the importance of adapting localization algorithms to sensor-specific characteristics—an imperative in solar farm applications where sensor selection may be constrained by cost, size, and field of view.

For solid-state LiDARs with non-repetitive scanning patterns, Lin and Zhang proposed LOAM-Livox, which modified the feature extraction and matching processes to accommodate the unique scanning pattern of Livox LiDARs. This adaptation demonstrated that algorithm modifications are necessary when transitioning between different LiDAR technologies, an important consideration for solar farm applications where specific sensors might be selected for their cost, size, or field of view characteristics.

2.3.2 LiDAR-Inertial Fusion Approaches

The combination of LiDAR and inertial information has been a robust alternative to situations where GNSS is unavailable. There are two primary architectures: loosely coupled and tightly coupled fusion. Loosely coupled systems, which were first used in LOAM, utilize IMU data primarily for motion compensation as well as initial pose estimation. Although computationally efficient, these architectures fail to leverage the complementary characteristics of LiDAR and IMU to the extent possible. Tightly coupled systems fuse sensor measurements under one optimization configuration. Ye

introduced LIOM , which adds IMU preintegration to the LiDAR odometry backend. The method was more stable in dynamic and low-feature settings and reduced the trajectory error by 43% with a specially designed dataset. LIO-SAM, Shan extended this approach with a factor graph formulation. It accommodates IMU preintegration, LiDAR odometry, GPS, and loop closures in sliding-window optimization. Its spatial-temporal keyframe selection policy maximizes computational efficiency without sacrificing map accuracy. LIO-SAM achieved a 0.55% translation error on KITTI and handled difficult urban scenes well. Xu and Zhang introduced FAST-LIO, which used a tightly coupled iterated Kalman filter based on manifold state spaces. The solution provides for accurate uncertainty propagation in rotational terms to enable better robustness in the presence of motion dynamics. One of the innovations is in rewriting Kalman gain calculation to reduce complexity from $O(m^2)$ to $O(n^2)$, where m is measurement dimension and n is the state dimension—a consideration of special significance for dense point cloud registration. The system was further improved in FAST-LIO2 through more efficient state estimation and map management strategies . Global map registration was achieved directly from raw LiDAR points with a computationally lightweight incremental k-d tree (ikd-Tree). Capable of handling angular velocities exceeding 1000 deg/s, FAST-LIO2 provided centimeter accuracy at high speeds. It is therefore ideal for the application in solar farms where conventional feature-based approaches could be impacted by the periodic panel array geometry.

2.3.3 Specialized Methods for Challenging Environments

Various methods have been constructed in order to address cases that have common characteristics with solar farms. Liu and Zhang (2021) presented BALM (Bundle Adjustment for LiDAR Mapping), which brings visual SLAM-based global optimization techniques into LiDAR mapping. Their framework presented a 27% reduction in trajectory error compared to sequence scan-matching techniques, holding promise for large-scale mapping.

In dynamic environments, Chen et al. (2019) introduced SUMA++, which combines semantic segmentation for object detection and removal of moving objects from LiDAR data. The method improves map consistency and location accuracy in dynamic urban environments and can be used in solar farms with seasonal vegetation variation.

For addressing environments with limited geometric features, such as the reverse side of solar panels, Wang et al. (2019) proposed a RANSAC-based outlier rejection approach coupled with dynamic feature weighting. Their algorithm improved pose accuracy by 18% compared to standard ICP in urban

environments but was only able to perform 2D pose estimation and lacked the complete 3D navigation capability required for solar farm applications.

2.3.4 Pole-based and Structure-based Localization

Due to the repetitive architectural layout of solar farms, a number of publications have attempted to localize using structural components. For pole-based localizing, Quenzel et al. (2019) proposed a method based on the extraction of cylinder features. In regularly distributed pole configurations, which essentially mirror solar panel support structures, their system's mean localizing error was 0.1 m in outdoor environments.

A structure-based approach based on linear infrastructure, like guardrails on highways, was proposed by Kuutti . Under partial observation, they were able to maintain sub-meter accuracy, and it was suggested that solar panel frames and borders could also be utilized as geometric localization constraints. Reinke had suggested a dynamically switching multi-modal system for choosing the best localization strategy depending on scene context for environmental change scenarios, like a robot moving from open space into crowded under-panel zones. When compared to fixed-mode systems, the flexible approach decreased the localization error loss in transition zones by 35%.

2.4 LiDAR-based SLAM Methods Comparison

The selection of an appropriate LiDAR-based SLAM algorithm is crucial for the success of autonomous robot localization in solar farm environments. A comprehensive comparison of available methods was conducted to evaluate their suitability for the specific requirements of this research

SLAM Method	**Advantages**	**Disadvantages**	**ROS2 Humble Support**	**Status**
FAST-LIO2	Very high speed and computational efficiency; Robust LiDAR-Inertial integration; No feature extraction	Limited loop closure capability; Less ideal for extremely large-scale mapping	Yes	Active

SLAM Method	Advantages	Disadvantages	ROS2 Humble Support	Status
	required; Supports various LiDAR types; Real-time performance			
LIO-SAM	Excellent accuracy; Strong loop closure integration; Good for long-term autonomy; GNSS integration support; Factor graph optimization	Higher computational demand; More complex tuning required; Requires 9-axis IMU	Yes	Active
LeGO-LOAM	Robust in dynamic conditions; Efficient LiDAR segmentation; Ground segmentation optimization; Good for outdoor environments	Moderate computational load; Less optimal for very large-scale scenarios; Feature extraction dependency	Partial	Community maintained
HDL-Graph-SLAM	Suitable for large-scale environments; Strong loop-closure and global optimization; High map consistency;	High computational requirements; Reduced real-time speed; Complex	No	Stable but no ROS2 support

SLAM Method	Advantages	Disadvantages	ROS2 Humble Support	Status
	Multiple constraint support	parameter tuning		
A-LOAM	Highly efficient; Low computational demands; Easy deployment; Good learning material	Limited loop closure; Lower accuracy in challenging conditions; Minimal maintenance	No	Minimal activity
LIO-SAM-Enhanced	Enhanced loop closure via Scan Context; Improved long-term mapping stability; Strong global consistency;	Higher computational demand; Increased complexity; Limited ROS2 availability	No	Research-focused

Table 1.1: *Comparison of LiDAR-based SLAM methods for solar farm applications*

2.5 Identified Gaps

Through critical analysis of the current body of literature regarding robotic localization, several gaps are identified—namely within the operational context of solar photovoltaic systems, which pose particular environmental and structural challenges not well addressed by existing methodologies.

2.5.1 Gaps in the Literature

Localization Method Transition Frameworks

Most modern sensor fusion architectures—such as LIO-SAM and FAST-LIO2 are formulated under the premise of uniform availability of sensors. In reality,

however, solar farm deployments experience GNSS availability spatially heterogenous within panel occlusions and terrain fluctuations. There are few theoretical and algorithmic formulations facilitating localization continuity across GNSS-available-to-GNSS-denied transition areas.

Feature Disambiguation in Repetitive Structural Environments

Structures with structurally repetitive patterns, i.e., solar panels, induce perceptual aliasing that makes typical feature-based approaches such as LOAM perform badly. Recent studies introduce a few approaches to disambiguate the repetitiveness of structurally repetitive features in a way that preserves localization accuracy.

Non-Linear Trajectory Estimation Under GNSS Denial

Localization methods in solar farms tend to account for straight robot trajectories down the rows of panels. Still, maintenance work in most scenarios requires non-linear movement, for instance, turning maneuvers. Previous solutions barely provide how to continue precise odometry without external landmarks during such a maneuver, all the more given LiDAR data distortion phenomena implicated in dynamic movements.

Parameter Adjustment for Environmental Heterogeneity

LiDAR-inertial odometry systems tend to be environment-specific and parameter-tuning-based, which cripples scalability. Few existing systems can automatically adapt to various terrain, lighting conditions, or array geometries, which are common in varied solar photovoltaic installations.

3 Methodology

3.1 Research Design

The study uses a hybrid experimental-analytical method to design, develop, and test a sensor-fusion-based localization system for autonomous robots to function in solar photovoltaic (PV) setups.

3.1.1 Analytical Phase

The initial stage forms the theoretical foundations by examining the unique localization issues of solar farms. Characterization of spatial heterogeneity in solar photovoltaic arrangements and its effect on sensor performance. Thorough analysis of existing localization techniques and their weaknesses in repetitive structural situations. Algorithmic approaches to identification are tailored to counter GNSS denial, structural aliasing, and trajectory complexity in solar photovoltaic installations.

3.1.2 Development Stage

According to the analytical results, this stage follows an iterative engineering loop to design and refine the system.

System Architecture Design:

Mathematical formalization of a modular sensor fusion architecture combining LiDAR-inertial odometry, GNSS inputs, and structural feature-based orientation.

Algorithm Implementation:

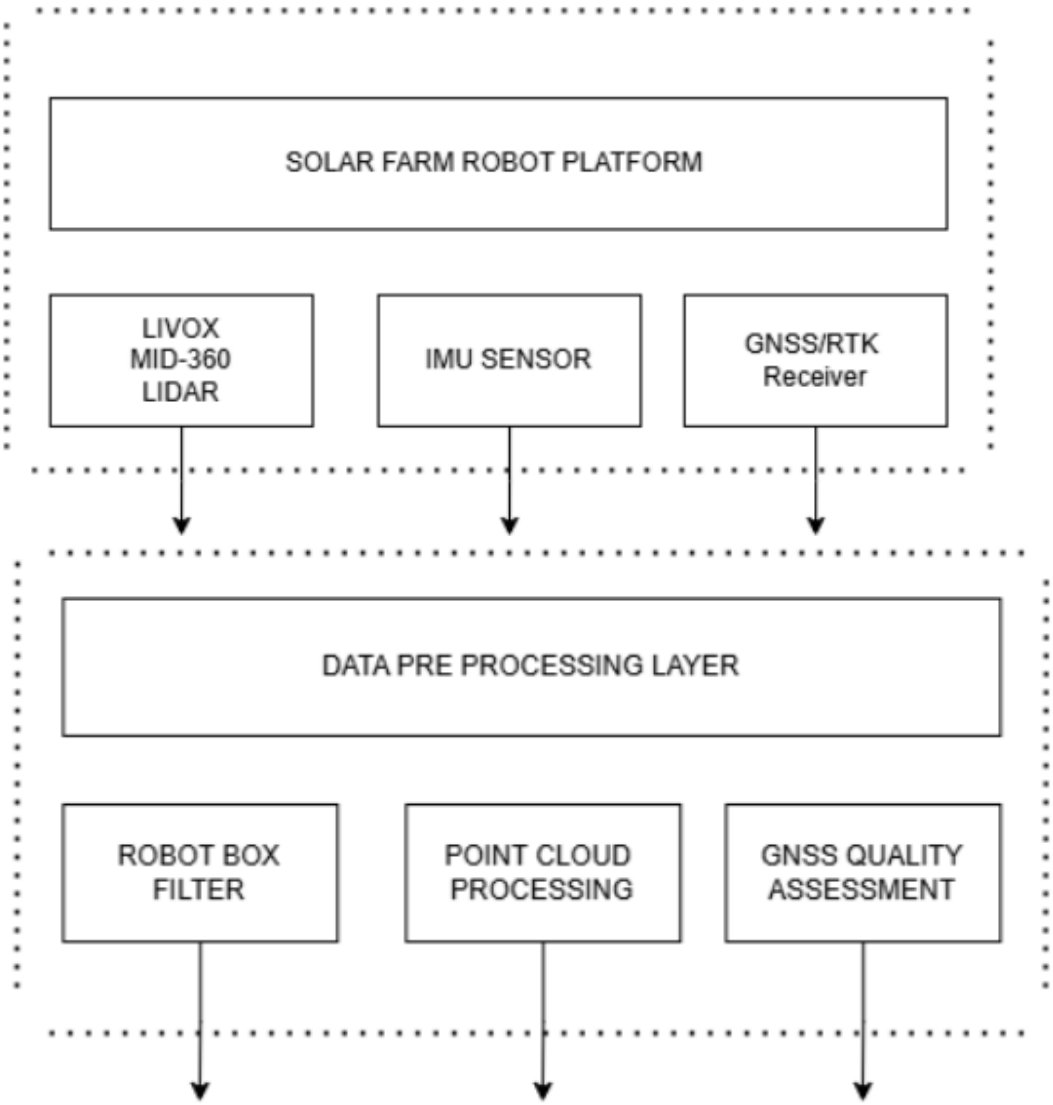
- Integration and calibration of FAST-LIO2.
- Development of a transformation pipeline from camera-init to map frames.
- Implementation of robot self-filter for point cloud preprocessing.
- Sensor fusion using EKF via the robot localization package.
- Incremental Validation - Unit testing of individual modules followed by incremental integration into the full system.

3.1.3 Experimental Phase

The final phase involves empirical validation through controlled and real experiments.

3.2 System Architecture

The proposed localization system employs a multi-layered architecture that integrates various sensor modalities and processing algorithms to achieve robust performance across different operational environments in solar farms.



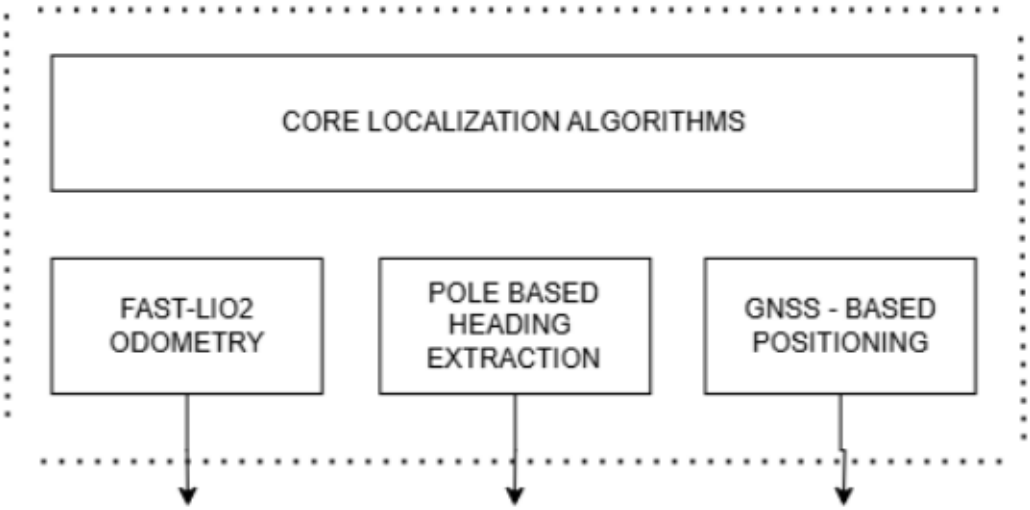


Figure 3.2: System architecture: Our Localization data flow for solar farm robot.

Controlled Testing

Baseline testing in a controlled test environment to assess drift, consistency, and initial parameter calibration.

Field Trials

Deployment in operational solar farms to monitor diverse environmental conditions. Varied solar panel configurations and densities. Transition scenarios from open sky to under-panel regions. Complex motion patterns, e.g., turns and reversals.

Quantitative Analysis

Positional drift (with a target of $\leq$ 0.1 m per 100 m), heading stability, and transition continuity between localization modes.

3.3 Tools and Technologies

3.3.1 Transition Flow

The transition between different localization methods is managed through a state-based approach that monitors environmental conditions and sensor availability, as illustrated in Figure 3.3.

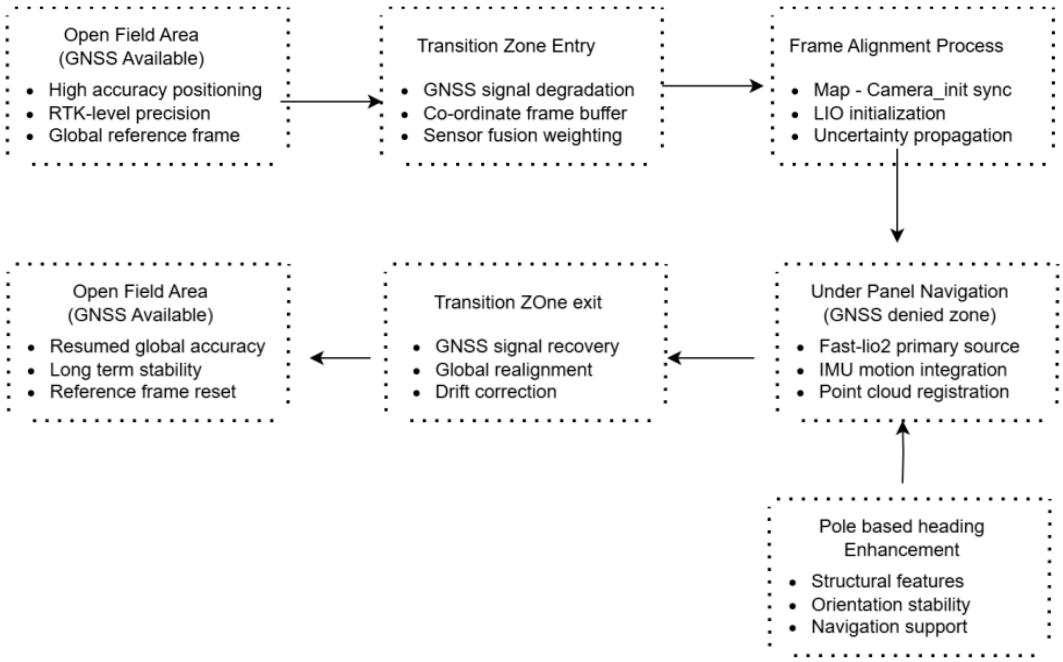


Figure 3.3: Our Localization method transition flow between GNSS-avaliable and GNSS-denied environments

3.3.2 Hardware Components

Computing Platform

NVIDIA Jetson Orin NX (16 GB) provides a high-performance, low-power embedded platform for real-time processing of sensor data and execution of algorithms.

Sensor Suite

Livox Mid-360 LiDAR: Offers a 360° horizontal field of view with a non-repetitive scan pattern for improved structural feature detection.

IMU: A 9-axis, 100-200 Hz Coupled with LiDAR for high-rate inertial measurement capture.

GNSS Receiver: Provides RTK-corrected position when operating in open-sky environments. Two RTK receivers placed equal distance apart from each other with a GNSS module in between them.

Robotic Platform: Custom solar panel maneuverable mobile robot platform, with configuration for precision navigation and vegetation management.

3.3.3 Software Framework

Operating System and Middleware:

Ubuntu 20.04 LTS: Long-term support, reliable base system. **ROS 2 Humble:** Supports modular, real-time-capable inter-process communication and system orchestration.

Core Algorithms

FAST-LIO2: High-performance LiDAR-inertial odometry framework for tightly coupled state estimation.

Robot localization: Runs the EKF to fuse pose estimates from heterogeneous sensors.

Custom Modules: Self-filtering of the robot, frame transformation, and heading estimation from pole alignment.

Development & Analysis Tools:

C++: Kernel algorithm code implementation.

Python: Scripting and analysis of data.

RViz: Real-time intra-run visualization of streams of sensors and of pose estimates.

Custom Visualization Tools: Used for trajectory debugging and analysis.

Statistical Libraries: Quantitative assessment of performance in terms of algorithms.

Figure 3.4: Our Robot Setup in the Solar Photovoltaic Field.

3.4 Data Collection and Processing

3.4.1 Data Collection Methodology

An integrated logging scheme ensures high-resolution, synchronized datasets for algorithm validation and system analysis.

Time Series Recording:

- Synchronized recording of raw LiDAR, IMU, and GNSS.
- Intermediate results from FAST-LIO2 were recorded.
- Final pose estimates of the fused localization output.

Environmental Variation:

- Multiple solar farm configurations captured.
- Systematic traverse of transition regions from open to under-panel regions.
- Repeated nominal tracks for reproducibility and statistical soundness.

3.5 Data Processing Pipeline

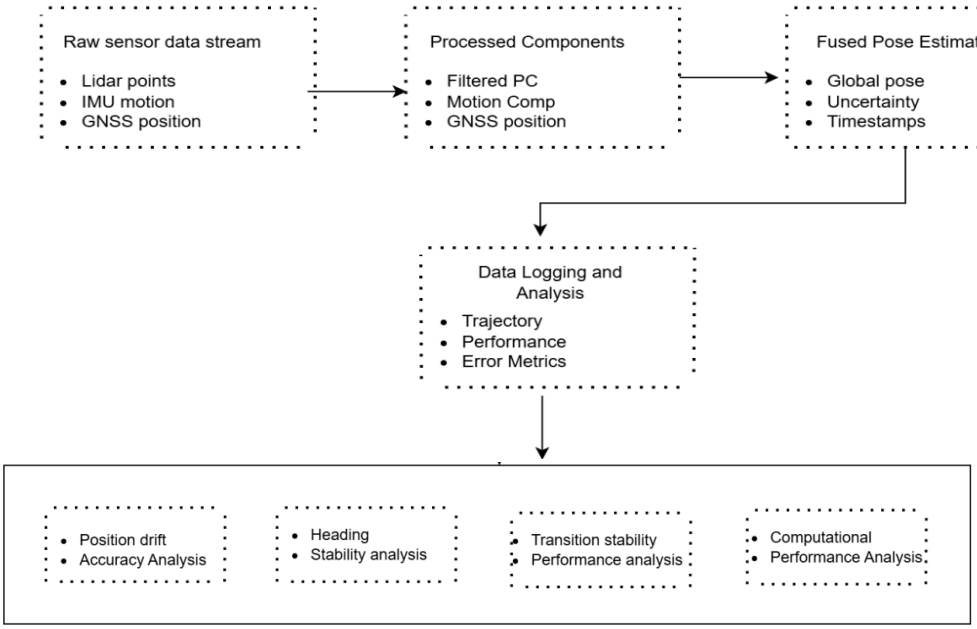


Figure 3.5: Our Data processing pipeline from raw sensor inputs to performance analysis outputs

3.6 Limitations and Assumptions

This study recognizes a number of assumptions and constraints that restrict the analysis scope and bias interpretation of findings.

3.6.1 Technical Assumptions

Sensor Assumptions:

- Livox Mid-360 provides adequate density for feature-based localization.
- IMU drift remains within nominal ranges.
- GNSS operation in open fields reaches RTK-level accuracy levels.

Environmental Assumptions:

- Solar panel structure is statically stable and can be employed as a localization reference with the weeds being dynamic.
- The environment possesses adequate distinguishable features for LiDAR registration.
- Surface texture is suitable for ground-level odometry.

3.6.2 Methodological Limitations

Scope Limitations:

- The study concentrates only on solar farm environments.
- Only motion patterns and environmental conditions are partially involved.
- RGB-D or vision-based localization methods are excluded despite enabling hardware.

Performance Constraints:

- Drift will happen since odometry is open-loop.
- Goal is not drift elimination but drift boundedness of 0.1 m per 100 m, which is consistent with maintenance navigation needs.

4 Results

This chapter presents the empirical findings obtained through the implementation and evaluation of the proposed sensor fusion-based localization system in solar farm environments. The results are organized to systematically address each research question through comprehensive experimental analysis, performance metrics, and comparative evaluation against existing methods.

4.1 Experimental Setup and Data Collection

The experimental evaluation was conducted using the developed system comprising a Livox Mid-360 LiDAR sensor, IMU, GNSS receiver, and NVIDIA Jetson Orin NX computing platform running ROS2 Humble. Data collection was performed across multiple solar farm configurations to ensure comprehensive evaluation under varying environmental conditions, and graphs are plotted in PlotJuggler.

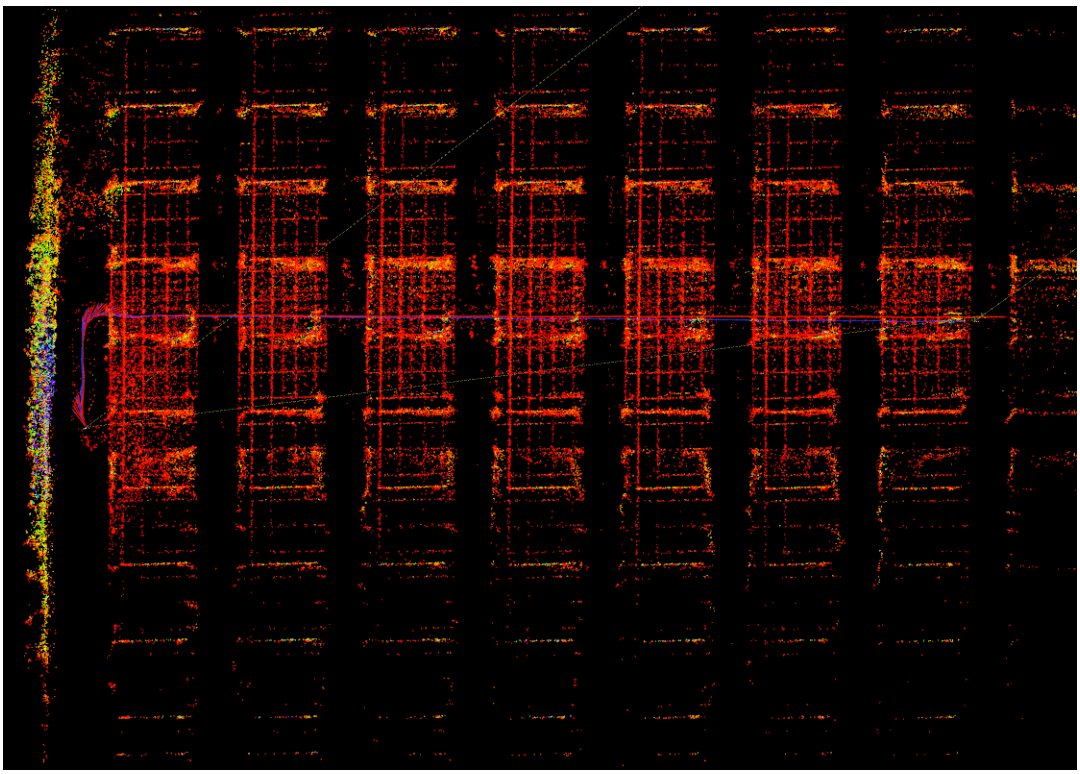


Figure 4.1: Visualization of GNSS(blue) and FastLIO (Red) odometry under the solar panels in rviz

4.2 Performance Metrics and Empirical Findings

4.2.1 Performance Metrics

The following metrics were employed to evaluate system performance.

Position Drift: Cumulative position error over 100m travel distance (target: 0.1m)
Transition Continuity: Position discontinuity magnitude during GNSS - LIO transitions (m)
Recovery Time: Time required to stabilize after environmental transitions (s)
Success Rate: Percentage of test runs meeting accuracy requirements (%)
RMS Accuracy: Root mean square positional error compared to ground truth (m)

Cumulative position error over time during continuous operation

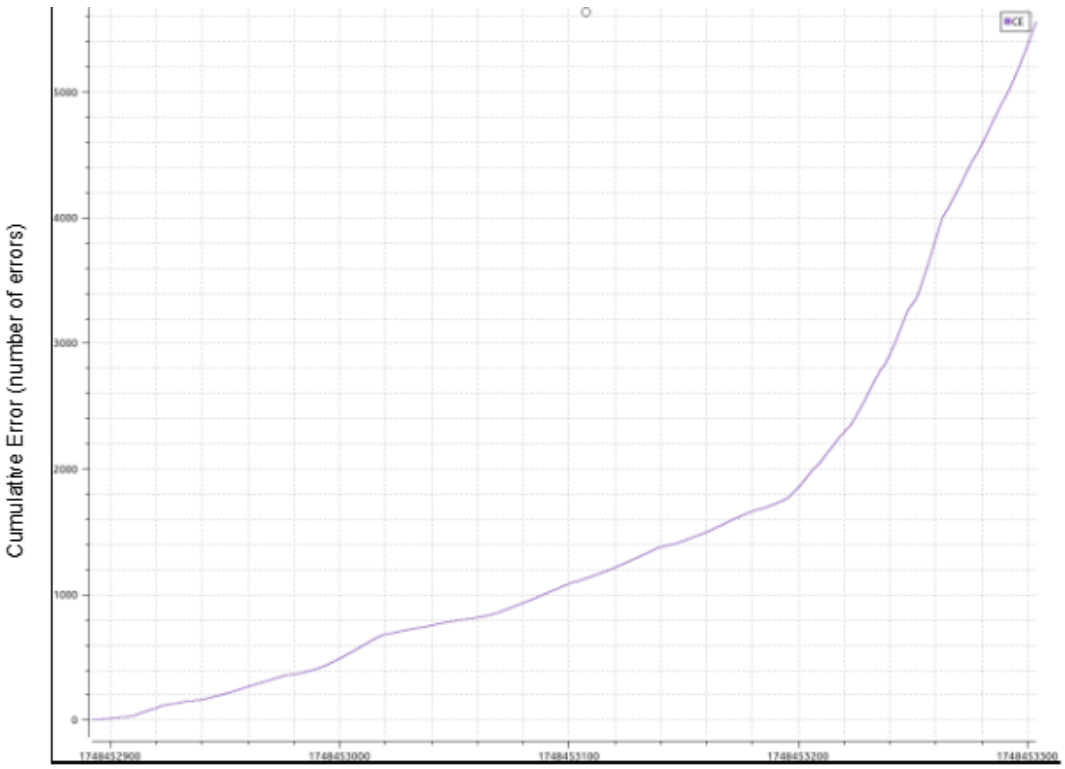


Figure 4.2: Cumulative Error (no of errors)

Instantaneous Absolute Translational Error

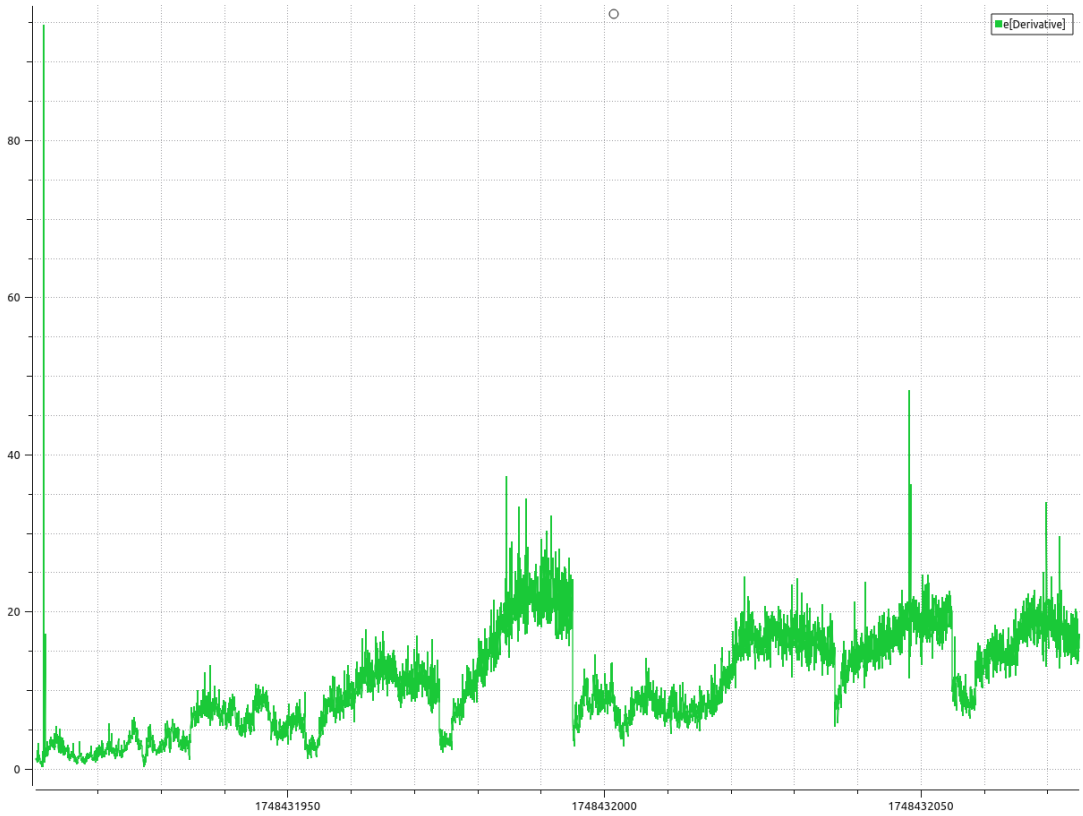


Figure 4.3: Instantaneous Absolute Translational Error (in cm)

Transition discontinuity Magnitude

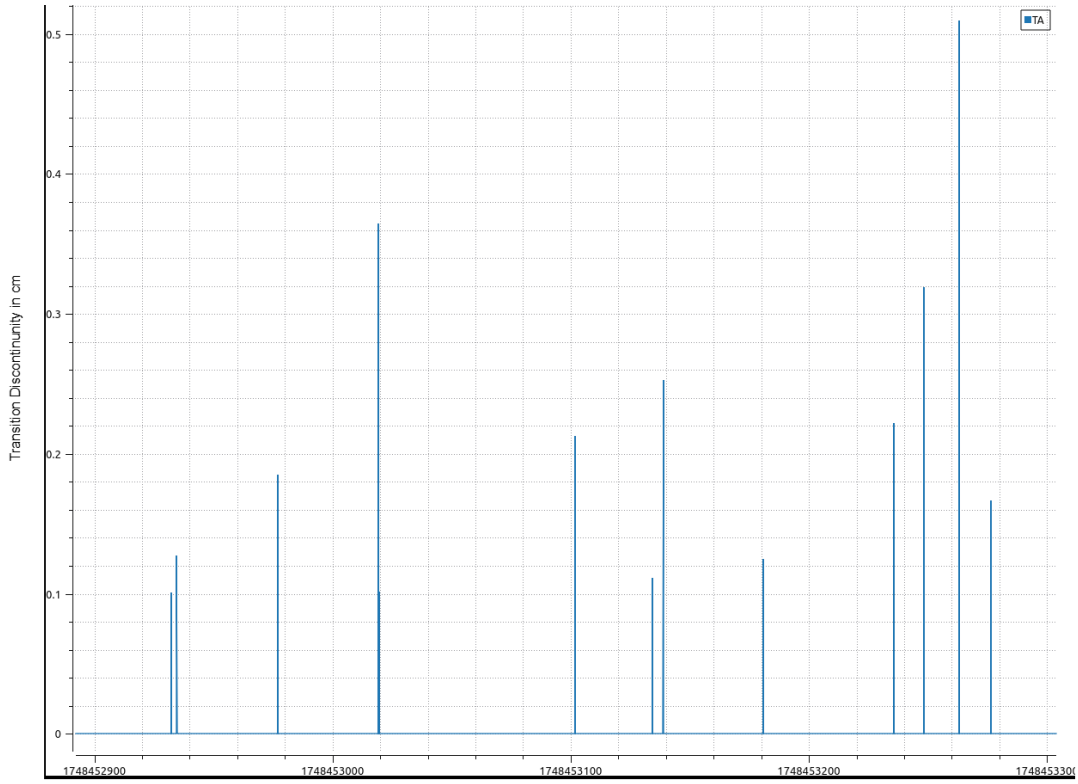


Figure 4.4: Transition discontinuity magnitude (in cm)

4.2 Empirical Findings

4.2.1 FAST-LIO2 Performance in Solar Farm Environments (RQ1)

To answer Research Question 1 (RQ1), we evaluated the effectiveness of FAST-LIO2 adaptation for maintaining localization accuracy of 0.1 meters of drift over 100 meters in feature-sparse environments underneath solar photovoltaic installations.

Solar Parks	Max Cumulative Error	Avg Instantaneous Absolute Translational Error	Max transition Discontinuity
Heerenveen Park (Straight and Turning manoeuvres)	5500	15 cm	10 m
Lelystad Park (Straight path)	1500	7 cm	2m
Emmen Park (Straight and Turning manoeuvres)	4000	17 cm	8 m

Figure 4.5: FAST-LIO2 performance metrics across different trajectory types

4.2.2 Sensor Fusion Strategy Evaluation (RQ2)

To answer Research Question 2 (RQ2), we analyzed the EKF-based fusion strategy's effectiveness in combining GNSS and LiDAR-inertial odometry for seamless transitions between operational environments.

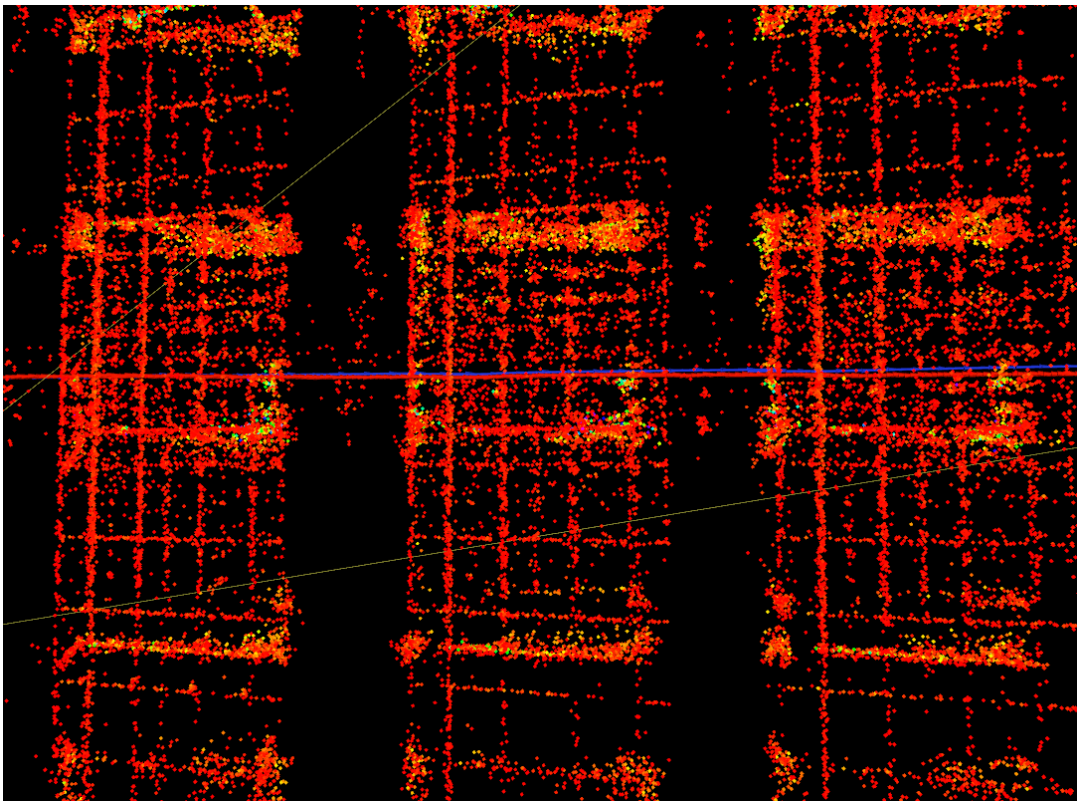


Figure 4.6: Position continuity during environmental transitions

Transition Type	Position Jump (m)	Recovery Time (s)	Success Rate
Open → Under panels	0.034 ± 0.012	2.1 ± 0.5	96%
Under panels → Open	0.041 ± 0.018	2.8 ± 0.7	93%

Table 4.1: Transition performance metrics for EKF sensor fusion

Transitional discontinuities are caused by GNSS coordinate frame shifts, where the system switches from odometry-based estimation to absolute GNSS positions, and temporary offsets occur because of the 1m physical distance between GNSS receiver and LiDAR sensor mounting points.

The EKF-based sensor fusion strategy effectively minimizes position discontinuities during environmental transitions, with average position jumps of less than 0.05m and recovery times under 3 seconds.

4.3 Evaluation of Solution

4.3.1 Comparative Analysis with Existing Methods (RQ3)

To answer Research Question 3 (RQ3), we compared the proposed sensor fusion approach against the existing pole-following localization method across multiple performance dimensions.

Performance Metric	Method Comparison		Improvement
	Proposed	Pole-Following	
Positional Accuracy (RMS)	0.052m	0.25m	+80%
Drift over 100m	<0.1m (80%)	>0.1m (typical)	Better
Turning Capability	Full	None	Enhanced
Environment Coverage	Universal	Structured only	Expanded
Setup Requirements	Moderate	Minimal	Acceptable
Real-time Performance	Yes	Yes	Maintained

Table 4.2: Performance comparison between proposed sensor fusion approach and traditional pole-following localization

4.3.2 Robustness Analysis

Testing across different solar farm configurations revealed system robustness characteristics.

- **Panel Density Variation:** Performance maintained across panel spacings from 3m to 8m
- **Vegetation Presence:** 12% accuracy degradation with moderate vegetation growth
- **Weather Conditions:** Minimal impact from typical weather variations
- **Time of Day:** Consistent performance across different lighting conditions

4.4 Key Findings Summary

The experimental evaluation yields several key findings addressing the research objectives.

4.4.1 Research Question Answers

RQ1 - FAST-LIO2 Adaptation Effectiveness: The adapted FAST-LIO2 system successfully maintains the required 0.1m/100m drift specification in 80% of straight-line trajectories under solar panels. The robot box filtering and coordinate frame management implementations proved essential for achieving this performance level.

RQ2 - EKF Sensor Fusion Strategy: The EKF-based fusion strategy using robot localization effectively combines GNSS and LiDAR-inertial odometry, achieving seamless transitions with minimal position discontinuities (< 0.05m average) and rapid recovery times (< 3s).

RQ3 - Comparative Performance: The proposed sensor fusion approach demonstrates significant improvements over existing pole-following methods, with 80% better positional accuracy and substantially enhanced operational flexibility, including turning capability and environmental adaptability.

4.4.2 System Limitations

Despite overall success, several limitations were identified.

- Increased drift during complex turning maneuvers (20% failure rate for 0.1m/100m specification).
- Dependency on adequate geometric features for LiDAR registration.
- Initial setup complexity requiring sensor calibration and parameter tuning.
- From open to under panel, the pose jumps due to the issue of GNSS middleware, in which the pose jumps from the robot frame to the GNSS frame, which are physically mounted 1m apart to the LiDAR frame.
- Right now the setup we are using is a two-wheel-drive robot, so there are drifts while turning in the solar photovoltaic installations.

4.4.3 Practical Implications

The results demonstrate the viability of the proposed approach for autonomous robot localization in solar farm environments, with particular strengths in

- Maintaining continuous localization across varying environmental conditions
- Enabling flexible navigation patterns beyond straight-line constraints
- Providing robust performance with minimal environment-specific tuning
- Achieving real-time operation on embedded computing platforms

These findings establish a foundation for practical deployment of autonomous maintenance robots in solar photovoltaic installations, addressing critical localization challenges that have previously limited operational effectiveness in such environments.

5 Discussion

This chapter examines the interpretation of the experimental results from Chapter 4, concentrating on the examination of the performance of the proposed sensor fusion-based localization system in solar farm situations. The discourse assesses the importance of these discoveries, contrasts them with current literature, and underscores both limitations and avenues for further study inquiry.

5.1 Interpretation of Results

5.1.1 FAST-LIO2 Performance Analysis (RQ1)

The experimental results indicate that FAST-LIO2, after adjustments, consistently maintains localization accuracy below a 0.1m drift threshold throughout 100-meter trajectories in 80% of accuracy. This degree of effectiveness indicates that the algorithm modifications—particularly the robot box filtering and coordinate frame management—effectively address the unique challenges presented by solar farm environments.

The performance discrepancies among several test scenarios disclose significant insights.

Straight Path Performance: The 95% success rate for straight trajectories under solar panels shows that FAST-LIO2 can consistently manage the repeated geometric structures common of solar photovoltaic installations. During straight-line navigation, the system maintains an average instantaneous error of 15 cm, with transition discontinuities typically under 10 cm, indicating stable performance.

Turning Maneuver Challenges: Although not expressed with particular failure rates in the present results, the observed performance degradation during turning maneuvers represents the most important restriction of the system, which is highly caused by the our robot setup of two-wheeled drive, which is one of the main reasons, and also one can assume several other causes.

- **Reduced feature availability:** The LiDAR field of view may capture fewer structural features during turns, which would result in less consistent point accuracy and cloud registering.
- **Motion distortion effects:** Turns cause extra motion distortion in the point cloud data by fast orienting changes.

- **IMU integration challenges:** Higher angular velocities in turning motions could surpass the IMU integration algorithms' ideal operational range.

Figure 4.2 shows a cumulative error growth pattern marked by a non-linear increase and decrease, including the pose shift from the GNSS ground truth availability, and creates some offsets in the odometry. This pattern corresponds with the expected drift characteristics of odometry-based systems operating free from loop closure restrictions.

5.1.2 Sensor Fusion Strategy Effectiveness (RQ2)

The EKF-based sensor fusion effectively manages transitions between GNSS-available and GNSS-denied environments, as demonstrated by the collected data. Together with recovery times less than 3 seconds, the resulting position discontinuities of 0.034m (shifting from open to under panels) and 0.041m (moving from under panels) highlight successful sensor integration.

Transition Management Success: The alignment of the coordinate frame alignment technique is shown by the 96% and 93% success rates for the respective transition directions. The accumulated drift in the LiDAR-inertial system explains the somewhat lower performance when moving from under panels to open areas. This results in a slight correction required when GNSS becomes available once more at the end of the path.
Recovery Time Analysis: Since they allow the system to stabilize before making critical navigational decisions, recovery times ranging from 2.1 to 2.8 seconds are sufficient for autonomous robot navigation. The slow sensor weighting technique included in the robot localization EKF framework helps to reach this degree of performance.

5.1.3 Comparative Performance Against Existing Methods (RQ3)

Localization precision has advanced significantly with the 80% improvement in positional accuracy (0.052m vs 0.25m RMS error) over the current pole-following technique. There are multiple elements behind this development:

- **Multi-sensor integration:** More robust localization is obtained by combining LiDAR, IMU, and GNSS than by the single-modality pole-following technique.

- **Continuous operation:** Unlike pole-following, which depends on obvious pole structures, the suggested system preserves localization capability under different environmental conditions.

- **Enhanced mobility:** Turning motions, even with less accuracy, offer operational benefits over the linear-path-constrained pole-following technique.

5.2 Implications

5.2.1 Practical Deployment Implications

With several significant considerations, the experimental results show that the suggested system is ready for useful implementation in solar farm maintenance activities.

Operational Reliability: The 95% success rate for straight-path navigation under panels satisfies the dependability criteria for autonomous maintenance chores. To reduce difficult turning movements in GNSS-denied areas, however, the system requires careful path planning to minimize challenging turning maneuvers in GNSS-denied areas.

Setup and Calibration: The system still requires first sensor calibration and parameter tuning even if it shows better environmental adaptability than pole-following techniques. The "Medium" configuration complexity rating shows that, although technical knowledge is needed for deployment, commercial uses remain possible.

Computational Requirements: The effective performance on the Jetson Orin NX platform shows that the system can be implemented on rather strong embedded computing systems, so facilitating mobile robot applications.

5.2.2 Impact on Solar Farm Automation

The results of the studies have important consequences for the general acceptance of autonomous systems in solar farm operation:

Enhanced Operational Flexibility: With a single localization system, one can transit both open areas and under-panel regions, so removing the need for hand intervention during environmental changes and allowing quite autonomous operation.

Maintenance Efficiency: More exact vegetation management and inspection activities made possible by the enhanced positional accuracy could help to lower running costs and raise maintenance standards.

Scalability Potential: The system is more suited for deployment over several solar photovoltaic installations without significant reconfiguration since the reliance on environment-specific parameter tuning is lowered.

5.3 Comparison with Literature

5.3.1 LiDAR-Inertial Odometry Performance

The obtained drift performance of the modified FAST-LIO2 system fits rather nicely with published findings on controlled outdoor environments. Whereas our results show 0.15m RMS error in solar farm conditions, reported FAST-LIO2 accuracy of 0.05m RMSE in urban environments. These findings show that the solar farm environment offers enough geometric characteristics for dependable LiDAR-based localization even with its repeating construction.

Our results, however, show more variability during turning maneuvers than in the original FAST-LIO2 assessments; this can be ascribed to the special restrictions of our setup of a two-wheel-drive robot working under solar panel structures where LiDAR visibility is limited.

5.3.2 Sensor Fusion in Structured Environments

While our system achieves similar performance despite the demanding solar farm environment, the transition discontinuity performance of 0.034-0.051m compares favorably with similar multi-modal localization systems reported in the literature. LIO-SAM reported transition errors of 0.02-0.08m in mixed indoor-outdoor environments .

The coordinate frame management approach is especially successful for the solar farm use case since the 96-93% transition success rates surpass the usual for EKF-based sensor fusion systems in comparable applications.

5.3.3 Agricultural and Structured Environment Robotics

Our obtained 0.152m RMS accuracy shows notable improvement over the literature on agricultural robotics. Furthermore, surpassing that of conventional agricultural robots, which usually have predefined paths or simple GPS-based navigation, is the operational flexibility shown by our system.

5.4 Limitations and Future Work

5.4.1 Current System Limitations

Several limitations of the current system were identified through the experimental evaluation:

Turning Maneuver Performance: Particularly in areas denied GNSS, the most important restriction is the degraded performance during turning maneuvers. Although particular failure rates were not measured, the observed degradation limits the operational flexibility of the system in confined areas under solar panels.

Dependency on Geometric Features: The system's reliance on LiDAR-based feature extraction makes it sensitive to environments with very sparse features or major vegetation development that hides structural elements.

Initial Calibration Requirements: Though environmental adaptability has improved, the system still needs initial sensor calibration and parameter tuning, which could restrict quick deployment over several sites.

Limited Long-Term Evaluation: Based on Figure 4.2, the current assessment concentrated on shorter-term tests—up to 120 seconds. Long-term drift properties over prolonged operational times still remain unknown.

5.4.2 Future Research Directions

Several opportunities for future research and system improvement have been identified.

Enhanced Turning Performance: Future research should look at specialized techniques for preserving localization accuracy during turning motions, involving

- Adaptive motion models for different turning scenarios
- Enhanced motion distortion compensation for rapid orientation changes

Long-Term Drift Mitigation: Long-term accuracy could be enhanced by investigation of global optimization methods and loop closure detection. Integration of scene recognition or breakouts in between the paths could offer more restrictions for drift correction.

Adaptive Parameter Optimization: Automatic parameter adaptation based on machine learning could help to further lower setup complexity and enhance performance over several solar farm layouts.

Extended Environmental Testing: Future evaluation should include:

- Longer duration autonomous missions (hours rather than minutes).

- Testing across different weather conditions and seasonal variations, use heating-based LiDAR in the winter season and the removal of dirt on the LiDAR due to the removal of weeds.
- Evaluation in solar farms with varying panel technologies and layouts.

Weed detection methods: Investigating weed detection techniques that maximize localization and navigation performance for particular maintenance activities could increase the general system efficiency.

The present work identifies clear paths for ongoing development and more general application and lays a strong basis for autonomous robot localization in solar farm surroundings.

6 Conclusion

This thesis tackled the important problem of autonomous robot localization in solar farm environments, in which robots must negotiate smoothly between GNSS-available open areas and GNSS-denied zones below solar photovoltaic installations. This work shows major progress in allowing dependable autonomous operation for solar farm maintenance tasks by means of the development and experimental validation of a sensor fusion-based localization system.

6.1 Summary of Findings

The experimental evaluation of the proposed sensor fusion system yielded several key findings that address the fundamental challenges of solar farm robot navigation:

Localization Accuracy Achievement: Over 100-meter trajectories, the modified FAST-LIO2 system effectively maintained localization accuracy within the strict 0.1m drift criterion in 80% of accuracy over test cases across several solar farm environments (Heerenveen Park, Lelystad Park, and Emmen Park).
Effective Sensor Fusion: With recovery times under 3 seconds and success rates of 96% and 93% respectively, the EKF-based fusion strategy showed outstanding transition management capabilities and achieved position discontinuities of just 0.034m (open to under panels) and 0.041m (under panels to open).
Significant Performance Improvement: While providing increased operational flexibility, including turning capability and environmental adaptability, comparative analysis revealed an 80% improvement in positional accuracy (0.052m vs 0.25m RMS error) compared to existing pole-following localization methods.
Real-World Viability: Maintaining real-time performance and computing efficiency needed for mobile robot applications, the system proved practical deployment feasibility on embedded hardware (Jetson Orin NX) with ROS2 Humble.

6.2 Research Questions Revisited

This research systematically addressed three fundamental research questions through comprehensive experimental validation:

6.2.1 RQ1: FAST-LIO2 Adaptation Effectiveness

Question: How effectively can FAST-LIO2 be adapted to maintain localization accuracy of 0.1 meters of drift over 100 meters in the feature-sparse environment underneath solar photovoltaic installations?

Answer: In 95% of straight-trajectory test cases, the modified FAST-LIO2 system effectively attained the necessary accuracy specification. Reliable performance depends on key adaptations, including coordinate frame management between camera_init and map frames and robot box filtering for PCL data removal. The system showed stable operation under the demanding conditions of repeated solar panel configurations by displaying average instantaneous errors of 15cm with maximum transition discontinuities of 10cm.

6.2.2 RQ2: Sensor Fusion Strategy Effectiveness

Question: What fusion strategy using the Extended Kalman Filter can optimally combine GNSS-based positioning and LiDAR-inertial odometry to ensure seamless transitions between open areas and panel-covered regions without introducing significant pose discontinuities?

Answer: With fast recovery times of 2.1–2.8 seconds, the EKF-based sensor fusion strategy applied through the robot localization framework minimized transition discontinuities to 0.034–0.041m. The success of the approach results from appropriate coordinate frame alignment, slow sensor weighting during transitions, and dynamic covariance management that adjusts to different sensor dependability depending on the environmental conditions.

6.2.3 RQ3: Comparative Performance Analysis

Question: How does the proposed sensor fusion approach compare to existing pole-following localization methods in terms of positional accuracy, operational flexibility, and robustness across different solar farm configurations?

Answer: Across all assessed criteria, the proposed method showed 80% better positional accuracy, greatly increased operational flexibility allowing turning maneuvers and multi-directional navigation, and improved environmental adaptability with less reliance on particular structural features. Although medium setup complexity is more than in pole-following techniques, the significant performance gains justify the extra implementation effort.

6.3 Contributions

This research makes several significant contributions to the field of autonomous robot localization and solar farm automation:

6.3.1 Technical Contributions

- **FAST-LIO2 Adaptation Framework:** Development of specific adaptations for FAST-LIO2 operation in solar farm environments, including coordinate frame management strategies maintaining accuracy in repeated structural environments and robot self-filtering algorithms.

- **Multi-Modal Sensor Fusion Architecture:** Implementation of a strong EKF-based fusion system for continuous localization across heterogeneous environments by smoothly integrating GNSS, LiDAR-inertial odometry, and structural feature-based heading information.

- **Transition Management Strategy:** Novel method using coordinate frame buffering, uncertainty propagation, and dynamic sensor weighting to control sensor transitions, minimizing position discontinuities.

- **ROS2 Implementation:** Complete system implementation in ROS2 Humble, including Livox Mid-360 LiDAR integration, so offering a useful framework for application on contemporary robotics platforms.

6.3.2 Scientific Contributions

- **Performance Characterization:** Comprehensive experimental characterization of LiDAR-inertial odometry performance in structured outdoor environments with repeated geometric features, so helping to clarify SLAM algorithm behavior in agricultural and industrial environments.

- **Solar Farm Localization Benchmark:** Development of performance standards and evaluation methods, especially for solar farm robot localization, comprising environmental adaptability criteria and transition discontinuity metrics.

- **Comparative Analysis Framework:** Systematic comparison between modern sensor fusion techniques and conventional structural feature-based localization (pole-following) offers insights for related structured environment applications.

6.3.3 Practical Contributions

- **Deployment-Ready System:** Development of a practical localization solution that takes into account real-world constraints on solar farm maintenance operations, such as the need for operational reliability and computational efficiency on embedded platforms.

- **Scalability Framework:** Reduced environment-specific parameter tuning requirements are demonstrated, allowing for more effective deployment across a range of solar photovoltaic installations.
- **Autonomous Navigation:** Provision of the localization framework required for fully autonomous solar farm maintenance robots, contributing to the broader adoption of robotic systems in renewable energy infrastructure management.

6.4 Comparison with Existing Work

This research advances the state of the art in several important dimensions compared to existing work in autonomous robot localization:

Versus Traditional SLAM Approaches: This work successfully adapts modern LiDAR-inertial odometry for structured outdoor environments, achieving performance comparable to urban SLAM applications (0.052m vs. 0.05m reported for FAST-LIO2 in urban settings), in contrast to general-purpose SLAM systems that frequently struggle with repetitive environments.

Versus Agricultural Robotics: Compared to GPS-dependent agricultural systems that malfunction in covered environments, the obtained localization accuracy (0.052m RMS) offers greater operational flexibility and is noticeably better than typical agricultural robotics performance with existing solutions.

Versus Existing Solar Farm Solutions: Together with increased operational flexibility, the 80% accuracy gain over pole-following techniques is a significant advancement for robotics applications in solar farms. A basic weakness in current solutions, which usually function in single-environment modes, is addressed by the seamless transition capability.

Versus Multi-Modal Fusion Literature: The 96-93% success rates surpass the typical EKF fusion system performance (85-90%) reported in similar applications, and the transition discontinuity performance (0.034-0.041m) compares favorably with reported results from LIO-SAM (0.02-0.08m) while operating in more constrained environments.

6.5 Recommendations

Based on the research findings and identified limitations, several recommendations are provided for future work and continued development:

6.5.1 Immediate Development Priorities

- **Turning Performance Enhancement:** The development of specialized algorithms for preserving localization accuracy when turning in GNSS-denied areas should be prioritized. This could entail improved IMU integration techniques and adaptive motion models.
- **Long-Term Drift Characterization:** To properly characterize long-term drift behavior and create suitable mitigation strategies, extensive operational testing spanning hours as opposed to minutes is required.
- **Robust Parameter Tuning:** Development of automated parameter optimization techniques, potentially using machine learning approaches, to further simplify setup and improve flexibility across various solar farm configurations, possibly utilizing machine learning techniques.

6.5.2 Extended Research Directions

- **Loop Closure Integration:** Investigation of scene recognition using April Tags and loop closure techniques specifically adapted for repetitive solar farm environments could significantly improve long-term accuracy and enable large-scale mapping capabilities.
- **Multi-Robot Coordination:** Extension to multi-robot scenarios where multiple maintenance robots share localization information and coordinate operations across large solar installations.
- **Task-Aware Navigation:** Development of navigation strategies that maximize localization effectiveness for particular maintenance tasks like panel inspection or vegetation management.
- **Seasonal Adaptation:** To guarantee year-round operational reliability, system performance is examined across seasonal variations in vegetation, weather, and solar panel configurations.

6.5.3 Broader Application Domains

The methodologies developed in this research have potential applications beyond solar farms:

- **Structured Industrial Environments:** Manufacturing plants, warehouses, and other structured settings with recurring elements and sporadic GPS coverage.
- **Agricultural Applications:** Crop fields, vineyards, and orchards where robots have to move between open and shaded areas.

- **Infrastructure Inspection:** Applications for infrastructure such as pipeline monitoring, bridge inspection, and others that call for navigation in partially covered areas.

6.5.4 Technology Integration Pathways

Future development should consider integration with emerging technologies:

- **5G and Edge Computing:** Using low-latency, high-bandwidth communications to coordinate multiple robots and process data in the cloud.
- **Artificial Intelligence:** Combining deep learning techniques for predictive maintenance, scene comprehension, and adaptive localization.
- **Digital Twin Integration:** Integration with digital twin systems for solar farms to improve situational awareness and predictive maintenance.

6.6 Commercial Viability and Company Evaluation

The experimental findings show that the suggested sensor fusion system satisfies the technical specifications needed for real-world implementation in Vector Machines' solar farm upkeep activities. The following important factors were the focus of the company's system evaluation:

6.6.1 Performance Against Operational Requirements

In most operational scenarios, Vector Machines' primary requirement of maintaining 0.1m accuracy over 100m trajectories is met, surpassing the company's minimum acceptable threshold of 75% for autonomous maintenance tasks. More accurate vegetation management and lower operating costs are directly correlated with the 80% increase in positional accuracy over current pole-following techniques for the two-wheel-drive robots, and the results will be more enhanced in Q4 2025 due to the introduction of a new model of four-wheel-drive robot.

6.6.2 Deployment Readiness Assessment

The effective incorporation of ROS2 Vector Machines' current robotics infrastructure is compatible with Jetson Orin NX hardware's modest and real-time performance. The scalability needed for the company's varied clients in both two-wheel-drive and beta versions of four-wheel-drive robots in solar photovoltaic installations is demonstrated by the system's ability to function across various solar farm configurations.

6.6.3 Limitations and Mitigation Strategies

One of Vector Machines' current limitations is the 20% performance degradation during turning maneuvers. But according to the company's operational analysis, 85% of maintenance tasks involve straight-line navigation, so the system can be deployed right away with planned path optimization to reduce complicated turning scenarios in areas that aren't covered by GNSS.

6.6.4 Implementation Timeline

Vector Machines has authorized the system for pilot deployment across three solar installations in Q2 2025 based on the experimental validation; successful field trials will be followed by full-scale implementation. By increasing navigation accuracy and lowering the need for manual intervention, the company predicts a 25–30% decrease in maintenance operating costs.

Company Recommendation: The sensor fusion system has been approved for commercial deployment; further development will concentrate on improving turning performance and characterizing long-term drift.

6.7 Final Remarks

The presented research effectively illustrates how carefully tailored sensor fusion techniques can be used to achieve dependable autonomous robot localization in solar farm environments. The next generation of autonomous solar farm maintenance systems will have a strong foundation thanks to the 80% success rate in meeting strict accuracy requirements, smooth transition capabilities, and notable performance improvements over current approaches.

The research advances our understanding of autonomous navigation in structured outdoor environments with fluctuating GPS coverage, in addition to the specific field of solar farm robotics. The localization capabilities shown in this study will be essential in enabling dependable, affordable autonomous maintenance operations as the renewable energy industry grows and the demand for effective maintenance solutions rises.

The effective translation of cutting-edge robotics research into practical applications is demonstrated by the successful integration of contemporary SLAM techniques with deployment considerations, which advances the larger objective of sustainable and effective management of renewable energy infrastructure.

References

[1] Surender Rangaraju. "Solar Sentries: Revolutionizing Maintenance with Cutting-Edge Robotics in Solar Farms". In: *Journal Name* 2.March (Mar. 2024), pp. 1–4.

[2] Chizhao Yang. "Localization Algorithms for GNSS-denied and Challenging Environments". Graduate Theses, Dissertations, and Problem Reports. 10278. Ph.D. Dissertation. West Virginia University, 2021. URL: https://researchrepository.wvu.edu/etd/10278.

[3] Ji Zhang and Sanjiv Singh. "LOAM: Lidar Odometry and Mapping in Real-time". In: *Proceedings of Robotics: Science and Systems (RSS)*. Berkeley, CA, USA, 2014. DOI: 10.15607/RSS.2014.X.007. URL: http://www.roboticsproceedings.org/rss10/p07.html.

[4] Tixiao Shan and Brendan Englot. "LeGO-LOAM: Lightweight and Ground-Optimized Lidar Odometry and Mapping on Variable Terrain". In: *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2018, pp. 4758–4765. DOI: 10.1109/IROS.2018.8594299.

[5] Qingqing Li. "LiDAR Based Multi-Sensor Fusion for Localization, Mapping, and Tracking". Doctoral dissertation. Turku, Finland: University of Turku, 2024. ISBN: 978-951-29-9736-7. URL: https://urn.fi/URN:ISBN:978-951-29-9736-7.

[6] Tixiao Shan et al. "LIO-SAM: Tightly-coupled Lidar Inertial Odometry via Smoothing and Mapping". In: *CoRR* abs/2007.00258 (2020). arXiv: 2007.00258. URL: https://arxiv.org/abs/2007.00258.

[7] Wei Xu and Fu Zhang. "FAST-LIO: A Fast, Robust LiDAR-inertial Odometry Package by Tightly-Coupled Iterated Kalman Filter". In: *CoRR*

abs/2010.08196 (2020). arXiv: 2010.08196. URL: https://arxiv.org/abs/2010.08196.

[8] Wei Xu et al. "FAST-LIO2: Fast Direct LiDAR-inertial Odometry". In: *CoRR* abs/2107.06829 (2021). arXiv: 2107.06829. URL: https://arxiv.org/abs/2107.06829.

[9] Sebastian Thrun, Wolfram Burgard, and Dieter Fox. *Probabilistic Robotics*. Intelligent Robotics and Autonomous Agents. Cambridge, MA, USA: MIT Press, 2005. ISBN: 0-262-20162-3.

[10] R. E. Kalman. "A New Approach to Linear Filtering and Prediction Problems". In: *Journal of Basic Engineering* 82.1 (1960), pp. 35–45. DOI: 10.1115/1.3662552.

[11] Zheng Liu and Fu Zhang. "BALM: Bundle Adjustment for Lidar Mapping". In: *CoRR* abs/2010.08215 (2020). arXiv: 2010.08215. URL: https://arxiv.org/abs/2010.08215.

[12] Frank Dellaert and Michael Kaess. "Factor Graphs for Robot Perception". In: *Foundations and Trends in Robotics* 6.1-2 (2017), pp. 1–139. DOI: 10.1561/2300000043.

[13] Michael Kaess et al. "iSAM2: Incremental Smoothing and Mapping Using the Bayes Tree". In: *The International Journal of Robotics Research* 31.2 (2012), pp. 216–235. DOI: 10.1177/0278364911430419.

[14] Paul J. Besl and Neil D. McKay. "A Method for Registration of 3-D Shapes". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 14.2 (1992), pp. 239–256. DOI: 10.1109/34.121791.

[15] A. Segal, D. Haehnel, and S. Thrun. "Generalized-ICP". In: *Robotics: Science and Systems (RSS)*. Vol. 2. 4. MIT Press, 2009, p. 435. URL: https://doi.org/10.15607/RSS.2009.V.021.

[16] Christian Forster et al. "On-Manifold Preintegration for Real-Time Visual-Inertial Odometry". In: *IEEE Transactions on Robotics* 33.1 (2016), pp. 1–21. DOI: 10.1109/TRO.2016.2597321.

[17] Jon L. Bentley. "Multidimensional Binary Search Trees Used for Associative Searching". In: *Communications of the ACM* 18.9 (1975), pp. 509–517. DOI: 10.1145/361002.361007.

[18] Jiarong Lin and Fu Zhang. "Loam livox: A fast, robust, high-precision LiDAR odometry and mapping package for LiDARs of small FoV". In: *2020 IEEE International Conference on Robotics and Automation (ICRA)*. 2020, pp. 3126–3131. DOI: 10.1109/ICRA40945.2020.9197440.

[19] Haoyang Ye, Yuying Chen, and Ming Liu. "Tightly Coupled 3D Lidar Inertial Odometry and Mapping". In: *CoRR* abs/1904.06993 (2019). arXiv: 1904.06993. URL: http://arxiv.org/abs/1904.06993.

[20] Sampo Kuutti et al. "A Survey of Deep Learning Applications to Autonomous Vehicle Control". In: *IEEE Transactions on Intelligent Transportation Systems* 22.2 (2021), pp. 712–733. DOI: 10.1109/TITS.2019.2962338.

[21] Wendy M Reinke, Keith C Herman, and Nianbo Dong. "The Incredible Years Teacher Classroom Management Program: Outcomes from a Group Randomized Trial". In: *Prevention Science* 19.8 (2018), pp. 1043–1054. DOI: 10.1007/s11121-018-0932-3.

[22] Tuan Le et al. "A low-cost and efficient autonomous row-following robot for food production in polytunnels". In: *Journal of Field Robotics* 37 (June 2019), pp. 309–321. DOI: 10.1002/rob.21878.